MT Classes, Motihari

The Mathematics of Maybe

Class 9 Maths, Chapter 7: Introduction to Probability (NCERT Ganita Manjari, Part I)



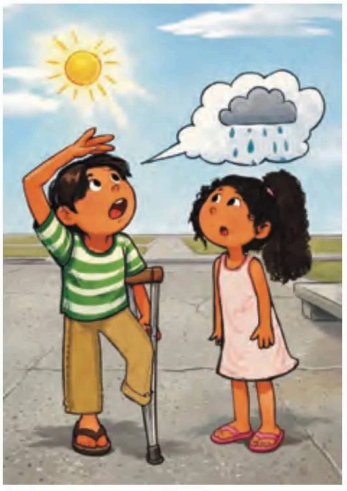
Coins, dice, spinners and sweets: learn to measure chance from 0 to 1. Flip a thousand coins in a second, build tree diagrams, drop paint on a target, and see every exercise solved step by step.

7.1 What is probability?

Probability is a kind of **measurement**. Length measures how long, area measures how much surface, and probability measures **how likely** an event is to happen.

You use it every day without noticing:

- 🌧️ **Is it going to rain today?**
- 🏑 **Will our school win the inter-school hockey match tomorrow?**
- 🎟️ **Will I be chosen in the monthly lucky draw to perform at the assembly?**



Sunny now, but will it rain later? Each friend reads the same sky differently.

These are **random events**. We know the **possible outcomes** (rain or no rain; win, draw or lose; one student gets picked) but we cannot know in advance which one will happen. There is an element of **chance** every time.

We describe chance with words like *impossible, certain, less likely, more likely* and *equally likely*. When a friend says, "The sun is shining, so it's unlikely to rain", that is a **subjective probability**: a guess based on how one person reads the evidence. This chapter is about measuring probability **objectively**, with data and counting.

7.1.1 What is randomness?

Randomness means you know all the things that *could* happen, but you cannot predict what *will* happen in one try.



Tossing a coin

Could be Heads or Tails. You can't be sure which in a single toss.



Rolling a die

Could be 1, 2, 3, 4, 5 or 6. You don't know which will show.

🔑 **Random experiment (trial)**

Something you can **repeat** (like tossing a coin) where the result may change every time and **cannot be known in advance**. All you can say is what *could* happen, not what *will* happen.

When you toss a coin, P(Heads) = $\frac{1}{2}$ and P(Tails) = $\frac{1}{2}$ because both are **equally likely**. Rain is random too: it depends on temperature, humidity, wind and pressure in such a sensitive way that no one can predict it with total certainty. Weather forecasters can only give a *probability* of rain from patterns in past data.

🤔 **Think and Reflect: Why is tossing a coin a fair way to decide which cricket team bats first?**

Because the toss is random and the coin is fair: Heads and Tails each have probability $\frac{1}{2}$. Neither captain can predict or control the result, so both teams have exactly the same chance of winning the toss.

7.1.2 The probability scale

Probability is measured on a scale from **0** to **1**, like a number line.

0	< ½	½	> ½	1
Impossible	Less likely	Even chance	More likely	Certain

- P = **0.75** means a 75% chance: *more likely than not*.
- P = **0.5** means 50%: *equally likely* to happen or not.
- P = **0**: impossible (winning a match without playing). P = **1**: certain.
- Most events have a probability strictly between 0 and 1.

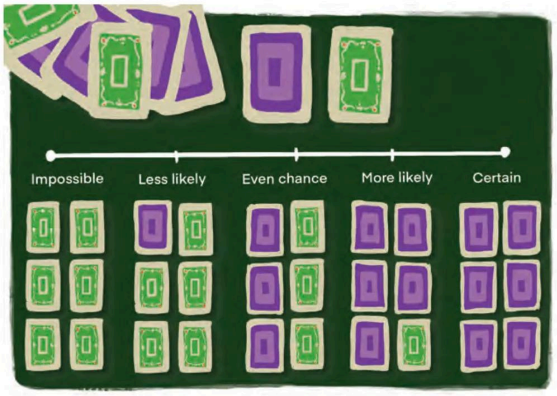


Fig. 7.1 Probability of picking a purple card from a deck of six, from impossible to certain.

Event	What it means
Getting a number greater than 6 on a die	**Impossible:** a die only has 1 to 6.
Rolling a 3 on a standard die	**Less likely:** but not impossible, one face is 3.
Flipping a coin and getting Heads	**Even chance:** Heads and Tails are equally likely.
Drawing any number from 2 to 10 from a deck of 52 cards	**More likely:** 36 of the 52 cards are numbered 2 to 10.
Choosing a red sweet from a bag of all red sweets	**Certain:** all the sweets are red.

Ex 7.1

Exercise Set 7.1: solutions

Tap a question to see the solution.

1 **Rank each event from 0 (impossible) to 1 (certain) and give reasons: (i) next Monday comes after Sunday (ii) snow in Mumbai in July (iii) an elephant walks through your classroom today (iv) you greet at least one friend at school tomorrow.**

Event	Label	Reason
(i) Monday after Sunday	Certain (1)	The days of the week always follow the same order.
(ii) Snow in Mumbai in July	Impossible (0)	Mumbai is hot and humid, and July is the monsoon season. It has never snowed there.
(iii) Elephant in your classroom	Less likely (almost 0)	Not strictly impossible, but extremely rare in a school.
(iv) Greet a friend tomorrow	More likely (close to 1)	You meet friends almost every school day, unless you are absent or it is a holiday.

Your reasons may differ slightly. What matters is that you justify your label.

7.2 Measuring probability objectively

There are two main ways to estimate the probability of an event objectively.

🧪

1. From experience: experimental probability

Collect data by repeating an experiment many times, or use past records. Probability = **relative frequency** of the event.

🧠

2. By reasoning: theoretical probability

Assume all outcomes are **equally likely**, then count favourable outcomes out of all possible outcomes. No experiment needed.

7.2.1 Experimental probability

In an experiment, each possible result is an **outcome**. The set of all possible outcomes is the **sample space**, written inside curly brackets.



Fig. 7.2 Heads and tails of a coin. Sample space {H, T}



Fig. 7.3 A 6-sided die. Sample space {1, 2, 3, 4, 5, 6}

Experimental Probability = $\dfrac{\textbf{Number of times the event occurred}}{\textbf{Total number of trials}}$

Also called the **relative frequency** of the event.

Example 2

A die is rolled 50 times and lands on 4 exactly 8 times.

Experimental probability of rolling a 4 = $\frac{8}{50}$ = **0.16** or **16%**. We also say the relative frequency of rolling a 4 is 0.16.

Relative frequency describes what actually happened in real data, not what we predict in theory. It is used a lot in statistics.

Lab The coin and dice lab

Run your own experiment. Toss a coin or roll a die as many times as you like and watch the **experimental probability** settle down next to the **theoretical probability** (dashed line). This is the **Law of Large Numbers**.

⚠️ Watch the streaks

Keep tossing one at a time. You will often see 4, 5 or even 6 Heads in a row. The coin has **no memory**: the chance of Tails next is still exactly $\frac{1}{2}$. Read about the Gambler's Fallacy in 7.2.3.

✨ Did you know?

The game of **Snakes and Ladders** comes from ancient India. It grew out of a dice game called **Jñān-Chaupaḍ**, used to teach moral lessons: each ladder stood for a virtue and each snake for a vice. Landing on a snake showed the result of bad behaviour; climbing a ladder showed the reward of good behaviour. Many versions have been found across India.

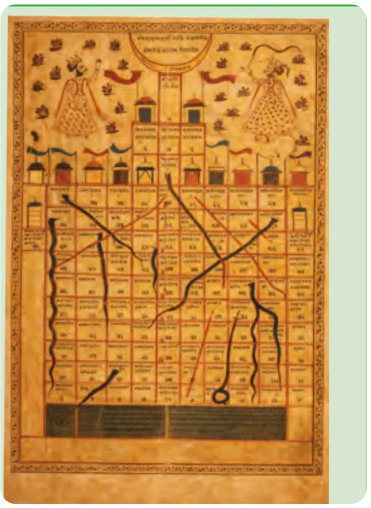


Fig. 7.4 A Jain Jñān-Chaupaḍ game on cloth, National Museum (India, 19th century)

7.2.2 Theoretical probability

Theoretical probability is what we **expect** in an ideal, perfectly fair situation where all outcomes are equally likely. No experiment is needed. We write it as P(Event).

$$P(E) = \frac{\text{Number of favourable outcomes}}{\text{Number of possible outcomes}}$$

"Favourable" means the outcomes where the event happens.

Example 3: Rolling a 4

Favourable outcomes = 1 (only 4)

Possible outcomes = 6 (1 to 6)

$P(4) = \frac{1}{6} = 0.1666\ldots \approx$ **0.167** or **16.7%**

Example 4: Letter B from PROBABILITY

Favourable = 2 (two Bs)

Possible = 11 letters

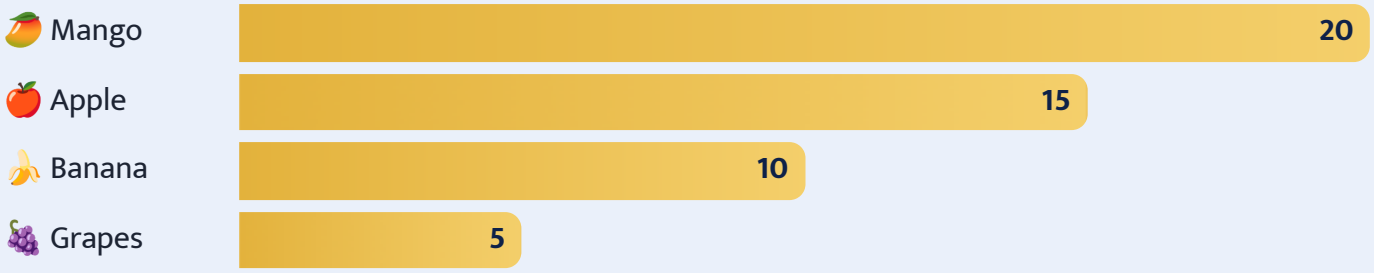
$P(B) = \frac{2}{11} = 0.1818\ldots \approx$ **0.182** or **18.2%**

7.2.3 Statistics, sampling and the Gambler's Fallacy

Businesses, insurance companies and scientists estimate probabilities from **collected data**.

Example 5: Favourite fruit

50 students in a class were asked their favourite fruit:



P(a randomly picked student likes mango) = $\frac{20}{50}$ = **0.4**, a 40% chance.

Scaling up: the whole school has 1500 students (the **population**). Our 50 students are a **sample**. Estimated mango lovers = 0.4 × 1500 = **600**. So buy about 600 mangoes.

📏 Sampling

A bigger sample (say 100 students) that is more **representative** (students from different classes) gives a more reliable estimate. Choosing the sample well is called **sampling**.

Experimental vs theoretical

	Experimental	Theoretical
Based on	Actual data from trials	Reasoning, assuming equally likely outcomes
Needs an experiment?	Yes	No
Formula	times event occurred ÷ total trials	favourable ÷ possible outcomes
Can change?	Yes, every new set of trials	No, it is fixed

📈 Law of Large Numbers

With few trials the two can be quite different. As the number of trials **increases**, the experimental probability gets **closer** to the theoretical probability. You saw this in the lab above.

🤔 Think and Reflect

If you rolled a 4 eight times in a row, the probability of rolling a 4 next is **still** $\frac{1}{6} \approx 0.16$ (for a fair die). Probability does not tell you what happens next; it predicts what happens **in the long run**.

🎰 The Gambler's Fallacy



Six Heads in a row. Surely Tails is "due" now? **No!** Each toss is a fresh start. P(Tails) on the next flip is still $\frac{1}{2}$. Believing that past results change the next one is called the **Gambler's Fallacy**.

Example 6: Snakes and Ladders

You rolled a 6 three times in a row and think, "No way I'll get a 6 again!" That is the Gambler's Fallacy. Each roll is **independent**, so P(6) is always $\frac{1}{6} \approx 0.166$ or 16.6%.

Key idea: randomness has no memory.

⚖️ Fair and unbiased

A **fair** (unbiased) coin is symmetrical, so there is no reason for it to fall on one side more often. A **random toss** means the coin falls freely without any interference.

Ex 7.2 Exercise Set 7.2: solutions

1 **Sample of 30 sweets: 10 red, 8 green, 7 yellow, 5 blue. (i) P(green)? (ii) If the bag has 600 sweets, how many are likely to be yellow?**

(i) P(green) = $\frac{8}{30} = \frac{4}{15} \approx$ **0.267**

(ii) P(yellow) = $\frac{7}{30}$. Expected yellow = $\frac{7}{30} \times 600 =$ **140 sweets**.

2 **Sample of 40 students: Science 14, Arts 11, Sports 9, Debate 6. School has 800 students. (i) P(Arts)? (ii) Estimate Sports club fans in the school.**

(i) P(Arts) = $\frac{11}{40}$ = **0.275**

(ii) $\frac{9}{40}$ × 800 = **180 students**.

3 **Toss a coin 20 times. (i) Heads? (ii) Tails? (iii) Experimental P(Heads)? (iv) P(Tails) on one more toss?**

Your answer depends on your own tosses. A sample result: 11 Heads and 9 Tails.

(i) 11 (ii) 9 (iii) Experimental P(H) = $\frac{11}{20}$ = **0.55**

(iv) P(Tails) = $\frac{1}{2}$, whatever happened before. The coin has no memory.

Use the Coin Lab above with "× 10" twice to do this experiment on screen.

4 **Toss a paper cup 100 times. Record bottom, top or side. Assign experimental probabilities.**

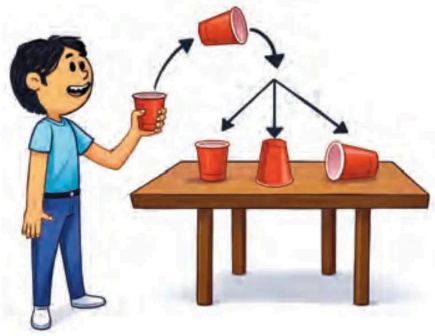


Fig. 7.5 Paper cup landing positions: bottom, top and side.

Sample record: bottom 18, top 24, side 58 (total 100).

P(bottom) = $\frac{18}{100}$ = 0.18, P(top) = $\frac{24}{100}$ = 0.24, P(side) = $\frac{58}{100}$ = 0.58.

The three probabilities always add up to 1. The outcomes are **not equally likely**, so we can only find these probabilities by experiment, not by theory.

5 **Probability of getting an even number on a fair 6-sided die?**

Even numbers: {2, 4, 6}, 3 favourable out of 6. P = $\frac{3}{6}$ = $\frac{1}{2}$

– **6** **A die rolled 12 times shows '3' three times. (i) Experimental P(3)? (ii) Theoretical P(3)? (iii) Why different? What if you roll 60, 600 or 6000 times?**

(i) $\frac{3}{12} = \frac{1}{4} = \mathbf{0.25}$ (ii) $\mathbf{\frac{1}{6}} \approx 0.167$

(iii) 12 rolls is a very small number of trials, so chance plays a big part. With 60, 600 or 6000 rolls we expect about 10, 100 or 1000 threes, and the experimental probability will get closer and closer to $\frac{1}{6}$ (Law of Large Numbers).

7.3 Sample spaces and events

7.3.1 Sample space

The **sample space S** is the list of **all** possible outcomes. Each outcome is an **element** of S.

✅ Include **every** possible outcome.

🚫 List no outcome **twice**.

#️⃣ The number of elements is the **sample size n(S)**.

Experiment	Sample space S	n(S)
Will it rain tomorrow?	{Rain, No Rain}	2
A team plays a match	{Win, Lose, Draw}	3
Toss a coin	{H, T}	2
Roll a die	{1, 2, 3, 4, 5, 6}	6
Toss two coins	{HH, HT, TH, TT}	4

🤔 Think and Reflect

{Rain, No Rain} only answers "will it rain?". If we care how much, we need a more detailed sample space: {No Rain, Drizzle, Light Rain, Heavy Rain}. The sample space must match the question being asked.

7.3.2 Events

An **event** is one outcome or a group of outcomes we are interested in. **An event is a subset of the sample space.**

Two coins

HH | HT | TH | TT

E = “at least one Head” = {HH, HT, TH}

One die

1 | 2 | 3 | 4 | 5 | 6

E = “greater than 4” = {5, 6}

Fruit basket

🍎 Apple | 🍌 Banana | 🍊 Orange

E = “a yellow fruit” = {Banana}

Ex 7.3

Exercise Set 7.3: solutions

1 **A single 6-sided die is rolled. Total number of outcomes in the sample space?**

S = {1, 2, 3, 4, 5, 6}, so n(S) = **6**.

2 **Write the sample space: (i) rolling a die and tossing a coin together (ii) a random integer between −5 and +5 (iii) one ball from a box of 5 green and 7 red balls.**

(i) S = {1H, 1T, 2H, 2T, 3H, 3T, 4H, 4T, 5H, 5T, 6H, 6T}, n(S) = **12**.

(ii) S = {−5, −4, −3, −2, −1, 0, 1, 2, 3, 4, 5}, n(S) = **11**. (If “between” excludes −5 and +5, S = {−4, ..., 4} with 9 elements.)

(iii) By colour: S = {Green, Red}. If we name each ball: S = {G1, G2, G3, G4, G5, R1, R2, ..., R7}, n(S) = 12. Only the second list has equally likely outcomes, so P(green) = $\frac{5}{12}$.

3 **Snacks: Samosa, Pakora, Bhaji. Drinks: Chai, Lassi. (i) Sample space of snack-drink combinations. (ii) Event “Samosa as the snack”.**

	☕ Chai	🥛 Lassi
Samosa	(Samosa, Chai)	(Samosa, Lassi)
Pakora	(Pakora, Chai)	(Pakora, Lassi)
Bhaji	(Bhaji, Chai)	(Bhaji, Lassi)

(i) S has 3 × 2 = **6** outcomes (listed in the table).

(ii) E = {(Samosa, Chai), (Samosa, Lassi)}

7.4 Tree diagrams

A **tree diagram** lists all outcomes of a **multi-step experiment** (a series of independent trials, like tossing a coin twice or rolling a die three times). Each branch is one possible result, and each path from start to end is **one complete outcome**.

Example 7: Toss a coin twice

1. From a start point, draw a branch for each result of the first toss (H, T).
2. From each of those, draw two branches for the second toss.
3. Read the 4 paths: S = {HH, HT, TH, TT}.

Write the probability on each branch ($\frac{1}{2}$).

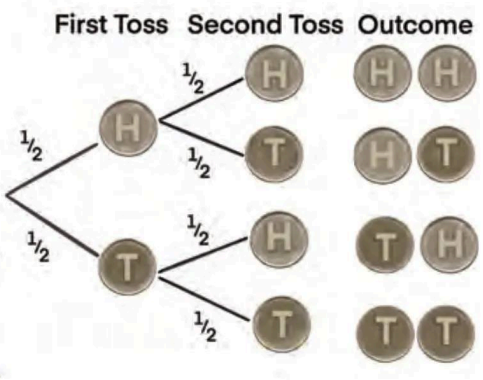


Fig. 7.6 Tree diagram for tossing a fair coin twice.

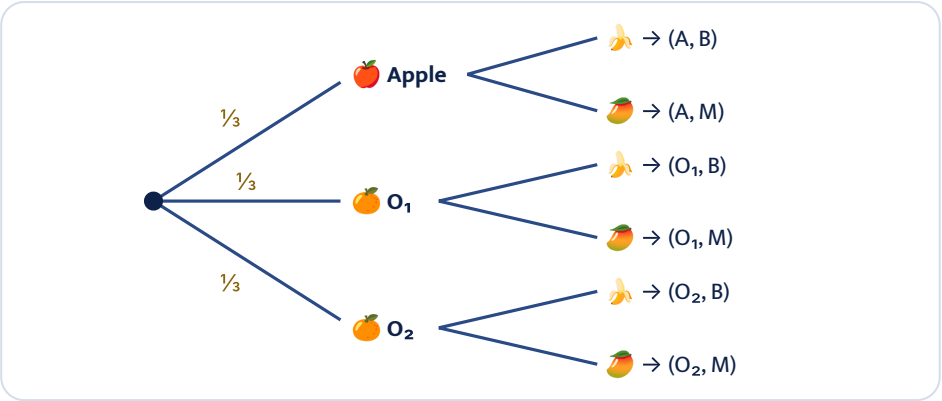
P(HH) = $\frac{1}{4}$ = **0.25** or 25% (1 favourable path out of 4). You can also multiply along the path: $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

🤔 **Think and Reflect: What is the probability of getting one Head and one Tail?**

Favourable paths: HT and TH, so 2 out of 4. P = $\frac{2}{4} = \frac{\mathbf{1}}{\mathbf{2}}$.

Ex 7.4 Exercise Set 7.4: solutions

1 **Basket A: 1 apple, 2 oranges. Basket B: 1 banana, 1 mango. Pick one fruit from each. (i) Tree diagram (ii) Sample space (iii) P(apple and banana)?**



(ii) S = {(A, B), (A, M), (O_1, B), (O_1, M), (O_2, B), (O_2, M)}: 6 equally likely outcomes. (By fruit type: Apple–Banana, Apple–Mango, Orange–Banana, Orange–Mango, but the orange pairs are twice as likely.)

(iii) Only (A, B) is favourable: P = $\mathbf{\frac{1}{6}}$. Along the path: $\frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$.

2 **A box has 3 red, 4 black, 2 green pens. You pick one and put it back, then your friend picks one. (i) Possible outcomes and tree diagram (ii) P(both the same colour)?**

(i) First pick: R ($\frac{3}{9}$), B ($\frac{4}{9}$), G ($\frac{2}{9}$). Because the pen is put back, the second pick has the same three branches. Colour outcomes: RR, RB, RG, BR, BB, BG, GR, GB, GG (9 paths). Counting each pen separately there are 9 × 9 = 81 equally likely outcomes.

(ii) Same colour = RR or BB or GG.

$P = \frac{3}{9}\times\frac{3}{9} + \frac{4}{9}\times\frac{4}{9} + \frac{2}{9}\times\frac{2}{9} = \frac{9}{81} + \frac{16}{81} + \frac{4}{81} = \mathbf{\frac{29}{81}} \approx 0.36$

End End-of-chapter exercises: solutions

1 **Fill in the blanks: (i) P(impossible event) = ___ (ii) The set of all possible outcomes is the ___ (iii) P(certain event) = ___ (iv) P(heads) for a fair coin = ___**

(i) **0** (ii) **sample space** (iii) **1** (iv) $\frac{1}{2}$ (0.5)

2 **In a survey of 50 students, 15 liked football. The ___ (frequency/relative frequency) is ___.**

The frequency is 15, and the **relative frequency** is $\frac{15}{50} = \frac{3}{10} = 0.3$.

3 **Which experiments have equally likely outcomes? (i) A car starts or does not start (ii) tossing a fair coin (iii) rolling a fair die (iv) a marble from 3 red and 7 blue (v) a baby is a boy or a girl.**

(i) **No.** A car in working order usually starts, so "starts" is far more likely.

(ii) **Yes.** H and T each $\frac{1}{2}$.

(iii) **Yes.** Each face $\frac{1}{6}$.

(iv) **No** (for colour). P(red) = $\frac{3}{10}$, P(blue) = $\frac{7}{10}$. Each single marble is equally likely, but the colours are not.

(v) **Nearly, but not exactly.** We often treat boy/girl as equally likely, but birth records show slightly more boys than girls are born.

4 **Write the sample space and find: (i) two coins, at least one head (ii) cards 1–10, even number (iii) a die, number > 4 (iv) 3 red, 2 blue, 1 green ball, not red (v) three coins, exactly two heads.**

(i) S = {HH, HT, TH, TT}; E = {HH, HT, TH}. P = $\frac{3}{4}$

(ii) S = {1, 2, ..., 10}; E = {2, 4, 6, 8, 10}. P = $\frac{5}{10} = \frac{1}{2}$

(iii) S = {1, ..., 6}; E = {5, 6}. P = $\frac{2}{6} = \frac{1}{3}$

(iv) S = {R1, R2, R3, B1, B2, G}; not red = {B1, B2, G}. P = $\frac{3}{6} = \frac{1}{2}$

(v) S = {HHH, HHT, HTH, HTT, THH, THT, TTH, TTT} (8 outcomes); E = {HHT, HTH, THH}. P = $\frac{3}{8}$

5 **A bag has 3 candies: strawberry, lemon, mint. P(strawberry)?**

S = {strawberry, lemon, mint}. P = $\frac{1}{3}$

6 2 shirts (red, blue) and 3 pants (jeans, khakis, shorts). List all outfits in a table.

	👖 Jeans	Khakis	🩳 Shorts
🔴 **Red shirt**	Red + Jeans	Red + Khakis	Red + Shorts
🔵 **Blue shirt**	Blue + Jeans	Blue + Khakis	Blue + Shorts

$2 \times 3 =$ **6 outfits**.

7 Tyre distances in 1000 cases: <4000 km: 20; 4001–9000: 210; 9001–14000: 325; >14000: 445. Find P(tyre lasts) (i) less than 4000 km (ii) between 4000 and 14000 km (iii) more than 14000 km.

Total = 20 + 210 + 325 + 445 = 1000 ✓

(i) $\frac{20}{1000} =$ **0.02**

(ii) $\frac{210+325}{1000} = \frac{535}{1000} =$ **0.535**

(iii) $\frac{445}{1000} =$ **0.445**

Check: 0.02 + 0.535 + 0.445 = 1.

8 Letters of PEACE on cards. (i) P(it is P, E or C)? (ii) P(not an E)?



5 cards: P, E, A, C, E.

(i) P, E, E, C are favourable: $P = \frac{4}{5}$

(ii) Not E: P, A, C. $P = \frac{3}{5}$

– **9*** **An arrow spins and stops on one of 1–8, equally likely. P(pointing at) (i) 8 (ii) odd (iii) more than 2 (iv) less than 9 (v) a multiple of 3?**

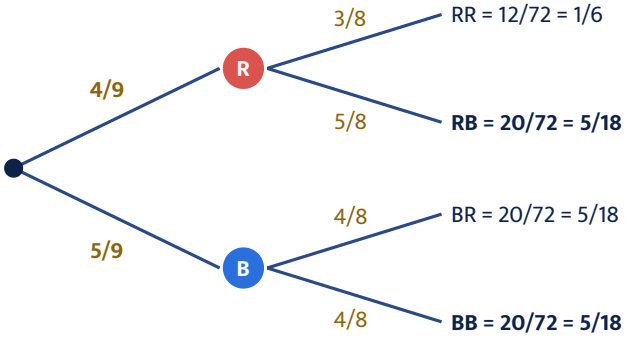


Fig. 7.7 The spinner. Play with it in the spinner game below.

(i) {8}: $\frac{1}{8}$ (ii) {1, 3, 5, 7}: $\frac{4}{8} = \frac{1}{2}$ (iii) {3, ..., 8}: $\frac{6}{8} = \frac{3}{4}$

(iv) All 8 numbers: $\frac{8}{8} = \mathbf{1}$ (certain) (v) {3, 6}: $\frac{2}{8} = \frac{1}{4}$

– **10*** **4 red and 5 blue balls. One ball is drawn and laid aside, then a second is drawn. Tree diagram; (i) P(red then blue) (ii) P(2 blue).**



After the first ball is laid aside, only 8 balls remain, so the second-draw probabilities change.

(i) P(R then B) = $\frac{4}{9} \times \frac{5}{8} = \frac{20}{72} = \mathbf{\frac{5}{18}}$

(ii) P(B, B) = $\frac{5}{9} \times \frac{4}{8} = \frac{20}{72} = \mathbf{\frac{5}{18}}$

Check: 12 + 20 + 20 + 20 = 72, so the four paths add up to 1.

11* **A pair of dice is thrown. Write an event with probability 0 and one with probability 1.**

Probability 0: "the sum is 13" (the largest sum is 6 + 6 = 12), or "the sum is 1".

Probability 1: "the sum is between 2 and 12" or "each die shows a number from 1 to 6".

Try both in the two-dice explorer in 7.3.

12* **(i) Two dice: sum a prime > 5 (ii) 4 red, 3 green, 2 blue; two balls without replacement: different colours (iii) three coins: first is heads and exactly two heads (iv) 4-digit number from 1, 2, 3, 4 (no repeats) is even (v) 3 MCQs with 4 options, guessing: exactly 2 correct.**

(i) n(S) = 36. Primes greater than 5 that can be a sum: 7 and 11.
Sum 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1): 6 ways. Sum 11: (5,6), (6,5): 2 ways.
$P = \frac{8}{36} = \mathbf{\frac{2}{9}}$

(ii) 9 balls. Number of ways to choose 2 balls = (9 × 8) ÷ 2 = 36.
Same colour: red pairs (4 × 3) ÷ 2 = 6, green pairs (3 × 2) ÷ 2 = 3, blue pairs 1. Total 10.
Different colours = 36 − 10 = 26. $P = \frac{26}{36} = \mathbf{\frac{13}{18}}$
Or by counting directly: R–G 4 × 3 = 12, R–B 4 × 2 = 8, G–B 3 × 2 = 6, total 26.

(iii) n(S) = 8. First coin H and exactly two heads: {HHT, HTH}. $P = \frac{2}{8} = \mathbf{\frac{1}{4}}$

(iv) Total numbers = 4 × 3 × 2 × 1 = 24. Even means last digit 2 or 4: for each choice of last digit, the other 3 digits can be arranged in 3 × 2 × 1 = 6 ways, so 12 even numbers.
$P = \frac{12}{24} = \mathbf{\frac{1}{2}}$

(v) Each question has 4 options, so n(S) = 4 × 4 × 4 = 64.
Choose which 2 questions are right: 3 ways (Q1Q2, Q1Q3, Q2Q3). Right answers: 1 way each; the wrong one: 3 ways. Favourable = 3 × 1 × 1 × 3 = 9.
$P = \mathbf{\frac{9}{64}} \approx 0.14$

13* **Balls numbered 1–4. Tree for (i) draw, replace, draw again (ii) draw, do not replace, draw again (iii) sizes of the two sample spaces.**

(i) With replacement: each of the 4 first branches splits into 4.
S = {11, 12, 13, 14, 21, 22, 23, 24, 31, 32, 33, 34, 41, 42, 43, 44}

(ii) Without replacement: each first branch splits into the 3 remaining numbers.
S = {12, 13, 14, 21, 23, 24, 31, 32, 34, 41, 42, 43}

(iii) 4 × 4 = **16** and 4 × 3 = **12**. (The 4 doubles 11, 22, 33, 44 are missing without replacement.)

14* Toss a coin and draw a card numbered 1–6. List the sample space.

S = {H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6}, n(S) = 2 × 6 = **12**.

15* Three coins are tossed and the number of heads recorded. Which list is the sample space: (i) {1, 2, 3} (ii) {0, 1, 2} (iii) {0, 1, 2, 3, 4} (iv) {0, 1, 2, 3}?

The number of heads can be 0 (TTT), 1, 2 or 3 (HHH). So **(iv) {0, 1, 2, 3}** is the sample space.

(i) misses 0 (all tails). (ii) misses 3 (all heads). (iii) includes 4, which is impossible with three coins.

16* A drop of dye falls at random on a 3 m × 2 m rectangle. P(it lands inside the circle of diameter 1 m)?

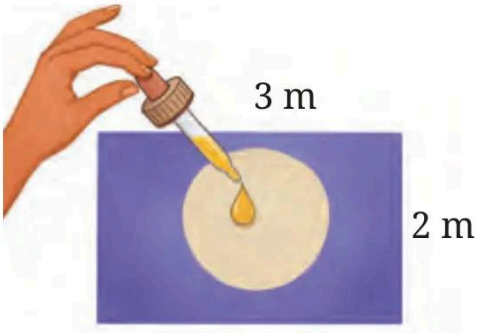


Fig. 7.8 Dye dropped at random on the rectangle.

When the drop is equally likely to land anywhere, probability = favourable **area** ÷ total **area**.

Rectangle = 3 × 2 = 6 m^2. Circle: r = 0.5 m, area = $\pi \times 0.5^2 = \frac{\pi}{4} \approx 0.785$ m^2.

$P = \frac{\pi}{4} \div 6 = \frac{\pi}{24} \approx$ **0.131** (using $\pi = \frac{22}{7}$: $\frac{11}{84} \approx 0.131$)

Test it with the dye-drop simulator below.

Sheet

Formula sheet and chapter summary

The scale

$0 \le P(E) \le 1$

0 = impossible, 1 = certain, ½ = even chance

Experimental

$$\frac{\text{times the event occurred}}{\text{total trials}}$$

Theoretical

$$P(A) = \frac{\text{favourable outcomes}}{\text{possible outcomes}}$$

Sample space

S = {all outcomes}, none repeated.
n(S) = sample size

Event

A subset of S: one or more outcomes we care about.

Tree diagram

Each path = one outcome. Multiply along a path, add the paths you want.

Law of Large Numbers

More trials ⟹ experimental gets closer to theoretical.

No memory

Past results never change the next toss (Gambler's Fallacy).

Handy counts

1 coin: 2 · 2 coins: 4 · 3 coins: 8
1 die: 6 · 2 dice: 36

📝 Chapter summary

- Probability measures the likelihood of an event, on a scale from 0 (impossible) to 1 (certain): $0 \leq P(E) \leq 1$.
- Experimental probability comes from doing an experiment: times the event occurred ÷ total trials.
- Theoretical probability assumes all outcomes are equally likely: favourable ÷ possible outcomes.
- A sample space S = {outcome_1, outcome_2, ..., outcome_n} lists every possible outcome of a random experiment.
- An event is one or a group of outcomes from the sample space.
- Tree diagrams list and visualise all outcomes and help us calculate probabilities.

Quiz Test yourself

Tap an answer. You get one try per question.

1. Which of these can NOT be a probability?

- 0
- 0.99
- **1.2**
- $\frac{3}{7}$

Probability is always between 0 and 1, so 1.2 is impossible.

2. A die is rolled. P(a number less than 3)?

- $\frac{1}{6}$
- **$\frac{1}{3}$**
- $\frac{1}{2}$
- $\frac{2}{3}$

{1, 2} out of 6: $\frac{2}{6} = \frac{1}{3}$.

3. A coin shows Heads 5 times in a row. P(Tails) on the next toss?

- Almost 1
- **$\frac{1}{2}$**
- $\frac{5}{6}$
- 0

The coin has no memory. Thinking otherwise is the Gambler's Fallacy.

4. How many outcomes when three coins are tossed?

- 3
- 6
- **8**
- 9

$2 \times 2 \times 2 = 8$.

5. A coin was tossed 40 times and showed Heads 22 times. Experimental P(Heads)?

- 0.5
- **0.55**
- 0.45
- 0.22

$\frac{22}{40} = 0.55$.

6. Two dice are rolled. P(a double)?

- $\frac{1}{36}$
- $\frac{1}{12}$
- **$\frac{1}{6}$**
- $\frac{1}{2}$

6 doubles out of 36 outcomes.

7. An event is a ___ of the sample space.

- multiple
- **subset**
- copy
- square

An event is made of some (or all, or none) of the outcomes in S.

8. P(choosing a vowel from the word MATHS)?

- **$\frac{1}{5}$**
- $\frac{2}{5}$
- $\frac{1}{4}$
- 0

Only A is a vowel: 1 out of 5 letters.

9. In a sample of 60 students, 15 walk to school. About how many of 1200 students walk?

- 150
- **300**
- 600
- 15

$\frac{15}{60}$ = 0.25 and 0.25 × 1200 = 300.

10. Which statement about the Law of Large Numbers is correct?

After many Heads, Tails becomes more likely	**With more trials, experimental probability gets closer to theoretical**
Theoretical probability changes with more trials	10 trials are always enough

Only the long-run proportion settles down; single tosses stay random.

MT Classes, S.K. Nagar, Motihari

Maths and Science coaching for Classes 8 to 10 (CBSE)

Notes based on NCERT Ganita Manjari (Grade 9 Mathematics, Part I), Chapter 7: The Mathematics of Maybe: Introduction to Probability. Diagrams are taken from the NCERT textbook chapter.